Improving Process Cycle Efficiency and On-Time Delivery Performance Using DMAIC

Lean Six Sigma Black Belt Candidate — Analytical Capstone Project

Dataset: DataCo Smart Supply Chain Dataset (Kaggle)

Methodology: Define • Measure • Analyze • Improve • Control (DMAIC)

Project Type: Historical Data Analysis and Proposed Process Improvement

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EXECUTIVE SUMMARY

This project applies to the Lean Six Sigma DMAIC methodology to evaluate supply chain delivery performance using the publicly available DataCo Smart Supply Chain dataset.

The analysis examined 180,519 order-line records, representing 65,752 unique customer orders. After excluding 2,855 canceled orders, 62,897 orders were eligible for delivery-performance evaluation. Of these, 36,048 were classified as late deliveries, resulting in a baseline late-delivery rate of 57.31% and an on-time or early delivery rate of 42.69%.

Microsoft Excel and Power Query were used for data preparation, KPI calculations, PivotTable analysis, and visualization. Pareto analysis identified Standard Class as the largest contributor to total late deliveries (41.60%), while First Class exhibited the highest late-delivery rate (100%).

An Ishikawa diagram and 5 Whys analysis were developed to examine potential contributing factors, including shipping schedule calibration, operational variability, and performance monitoring.

Three hypothetical improvement scenarios were evaluated, targeting late-delivery rates of 50%, 45%, and 40%. Under the proposed 45% target, approximately 7,744 late orders could be avoided over an equivalent order volume.

A proposed control plan was developed to monitor delivery reliability, shipping schedule variance, and mode-specific performance.

The study demonstrates practical application of DMAIC and data-driven process improvement techniques. However, the recommended improvements have not been implemented or experimentally validated. The findings represent historical observations and hypothetical improvement scenarios rather than verified operational results.

1. DEFINE PHASE

1.1 Project Background

Supply chain delivery performance is a critical component of customer satisfaction and operational efficiency. Variability in shipping schedules, fulfillment processes, and delivery execution can contribute to late deliveries and reduced service reliability.

This project applies the Lean Six Sigma DMAIC methodology to the publicly available DataCo Smart Supply Chain dataset to evaluate delivery performance, identify potential drivers of delays, and develop data-driven improvement recommendations.

The project is an analytical case study based on historical data. Proposed improvements will be evaluated through scenario analysis rather than actual operational implementation.

1.2 Project Charter

Problem Statement

Analysis of the DataCo Smart Supply Chain dataset identified 62,897 eligible customer orders, excluding canceled shipments. Of these, 36,048 orders were classified as late deliveries, representing a late-delivery rate of 57.31%. This indicates a significant delivery reliability problem that warrants systematic investigation using the DMAIC methodology.

Project Goal

To investigate the factors associated with late deliveries and evaluate proposed process improvements capable of reducing the late-delivery rate from 57.31% to 45% or below, representing an approximately 21.5% relative reduction.

Business Case

Improving on-time delivery performance could increase customer satisfaction, reduce operational inefficiencies, improve delivery predictability, and support better supply chain decision-making.

Project Scope

In Scope:

- Historical order-level delivery performance
- Shipping modes and scheduled versus actual shipping times
- Delivery delays and process variation
- Root cause analysis using available data
- Proposed improvement scenarios and control mechanisms

Out of Scope:

- Physical implementation of improvement recommendations
- Actual financial savings validation
- Real-time carrier performance monitoring

Primary KPI: Order-Level Late Delivery Rate

Secondary KPIs: On-Time Delivery Rate, Average Shipping Duration, Shipping Schedule Variance, and Shipping Mode Performance

Process Cycle Efficiency: To be assessed only if sufficient value-added and total lead-time information is available.

Expected Deliverables: DMAIC analytical report, baseline KPI analysis, Pareto chart, root cause analysis, proposed improvement scenarios, control plan, and GitHub portfolio documentation.

2. MEASURE PHASE

2.1 Dataset Preparation and Validation

The analysis utilized the publicly available DataCo Smart Supply Chain dataset, containing 180,519 order-line records. Microsoft Excel Power Query was used to prepare an order-level analytical dataset by selecting delivery-related variables and retaining one record per unique Order ID.

The resulting dataset contained 65,752 unique customer orders. A total of 2,855 canceled orders were excluded from the delivery-performance denominator, leaving 62,897 eligible orders.

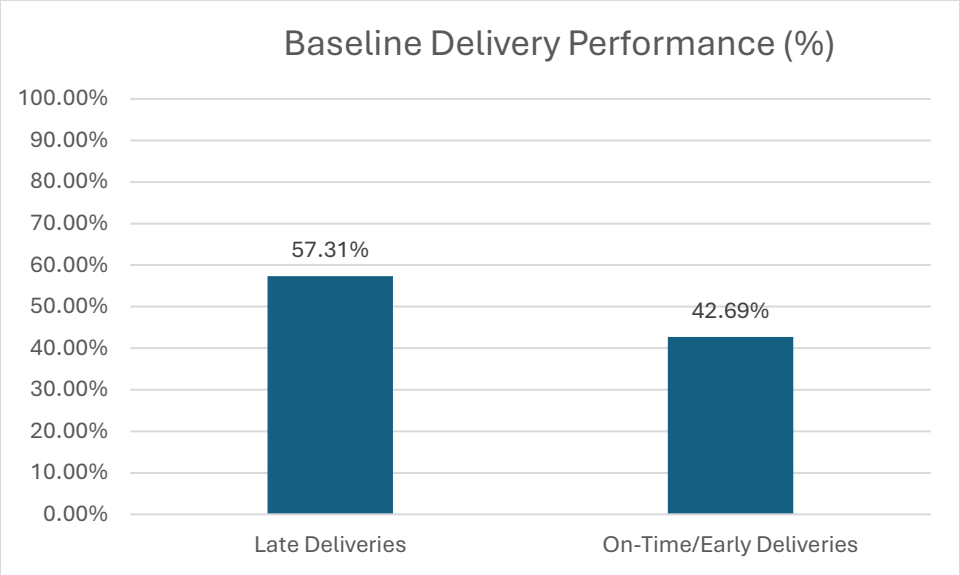
2.2 Baseline Delivery Performance

The baseline analysis identified 36,048 late deliveries and 26,849 on-time or early deliveries.

The calculated order-level late-delivery rate was **57.31%**, while the on-time delivery rate was **42.69%**.

These results establish the historical performance baseline against which proposed improvement scenarios will be evaluated.

Figure 1: Baseline Delivery Performance (%)



2.3 Baseline Interpretation

The findings indicate that late deliveries accounted for more than half of eligible orders in the historical dataset. This represents a substantial delivery reliability gap and provides justification for further investigation of shipping methods, scheduling practices, and process variation during the Analyze phase.

3. ANALYZE PHASE

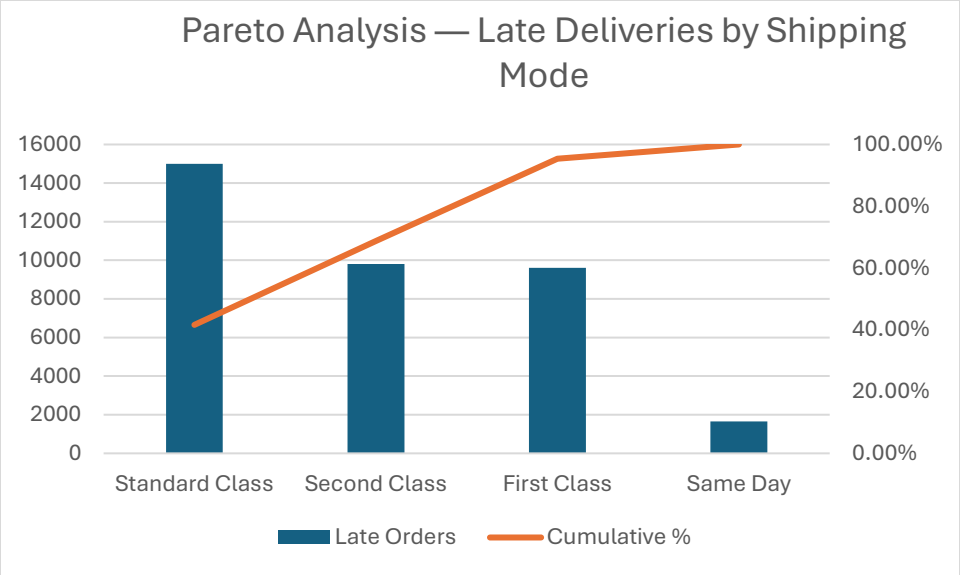
3.1 Pareto Analysis of Late Deliveries

A Pareto analysis was conducted to identify the shipping modes contributing the greatest number of late deliveries.

The analysis revealed that Standard Class accounted for 41.60% of late deliveries, followed by Second Class (27.19%), First Class (26.64%), and Same Day (4.57%).

Collectively, Standard Class, Second Class, and First Class accounted for approximately 95.43% of all late deliveries.

Figure 2: Pareto Analysis — Late Deliveries by Shipping Mode



3.2 Interpretation of Shipping Mode Performance

Although Standard Class contributed the largest absolute number of late deliveries, First Class recorded the highest late-delivery rate (100%), followed by Second Class (79.99%), Same Day (48.37%), and Standard Class (39.85%).

These results indicate substantial variation in delivery performance across shipping modes.

However, shipping-mode associations do not establish the underlying causes of late deliveries. Further investigation is required to determine whether scheduling policies, fulfillment processes, or other operational factors explain the observed differences.

3.3 Cause-and-Effect (Ishikawa) Analysis

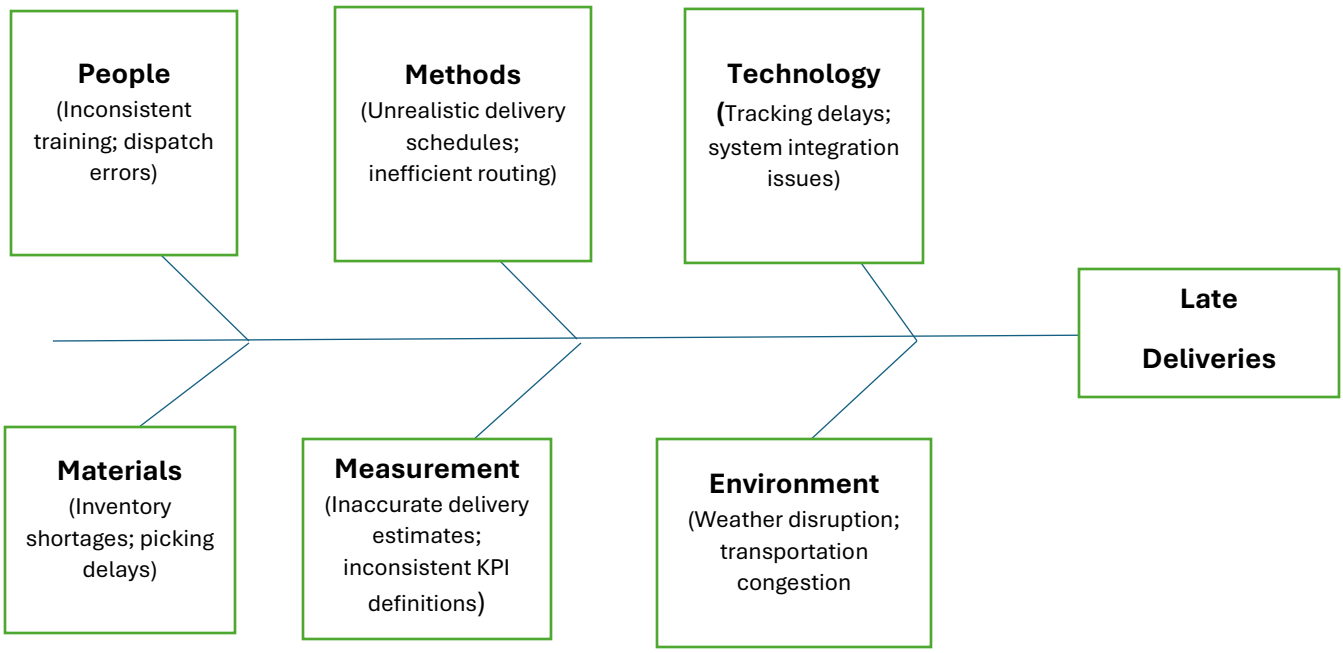


Figure 3: Cause-and-Effect (Ishikawa) Diagram — Potential Causes of Late Deliveries

The Ishikawa analysis identified six categories of potential contributors to late deliveries: People, Methods, Technology, Materials, Measurement, and Environment.

Possible causes include unrealistic delivery schedules, inefficient routing, inventory shortages, tracking-system limitations, and transportation disruptions.

These factors represent hypotheses requiring further operational investigation. The historical dataset provides evidence of differences in delivery performance across shipping modes but does not independently confirm the underlying operational causes.

3.4 Five Whys Analysis — Shipping Schedule Misalignment

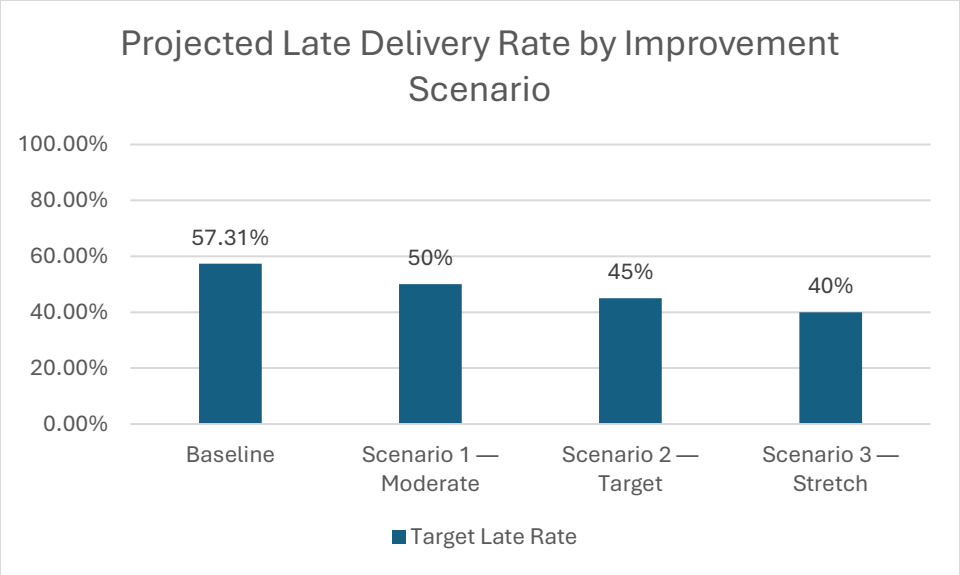
Level	Question	Potential Explanation
Problem	Why are orders delivered late?	Actual shipping durations may exceed scheduled durations.
Why 1	Why might actual durations exceed schedules?	Shipping schedules may not adequately reflect operational variability.
Why 2	Why might schedules overlook variability?	Different shipping modes may use insufficiently calibrated delivery targets.
Why 3	Why might delivery targets be poorly calibrated?	Historical performance may not be systematically incorporated into scheduling decisions.
Why 4	Why might historical performance be underused?	Performance monitoring and feedback processes may be inadequate.
Why 5	Why might monitoring be inadequate?	Clear accountability for reviewing and updating shipping targets may be missing.

3.5 Analyze Phase Summary

The Pareto analysis identified Standard Class as the largest contributor to total late deliveries (41.60%), while First Class exhibited the highest late-delivery rate (100%).

The Ishikawa diagram and 5 Whys analysis identified potential weaknesses in shipping schedule calibration, operational variability, and performance monitoring.

These findings provide a basis for developing targeted improvement scenarios. However, the identified operational causes remain hypotheses requiring further validation.



5. CONTROL PHASE

5.1 Proposed Process Control Plan

The Control phase establishes a proposed framework for sustaining delivery-performance improvements if the recommended solutions are implemented. The control plan defines key performance indicators, monitoring frequencies, responsible personnel, and corrective actions to address deviations from established performance targets.

As this project uses historical data, the control mechanisms presented are recommendations rather than implemented operational controls.

KPI	Proposed Target	Frequency	Responsible Role	Corrective Action
Late Delivery Rate	≤45%	Weekly	Logistics Manager	Investigate deviations and initiate corrective actions
On-Time Delivery Rate	≥55%	Weekly	Operations Manager	Review delayed orders and delivery processes
Shipping Schedule Variance	Reduce from baseline	Weekly	Transportation Supervisor	Review actual vs. scheduled durations
Shipping Mode Performance	Improve mode-specific baseline	Monthly	Supply Chain Analyst	Analyze trends and recommend adjustments

5.2 Control Phase Summary

The proposed control plan establishes a framework for monitoring delivery reliability and sustaining potential process improvements. Weekly reviews of delivery performance, shipping schedule variance, and corrective actions are recommended to identify deviations from established targets.

Monthly shipping-mode performance reviews would provide additional opportunities to evaluate operational trends and identify recurring delivery challenges.

As this project is based on historical data, the control plan has not been implemented. Future validation would require operational testing and post-implementation performance measurements.

6. CONCLUSION AND RECOMMENDATIONS

This Lean Six Sigma DMAIC case study evaluated delivery performance using the DataCo Smart Supply Chain dataset.

Analysis of 65,752 unique customer orders identified 62,897 eligible orders, of which 36,048 were classified as late deliveries. The resulting baseline late-delivery rate was 57.31%.

Pareto analysis identified Standard Class as the largest contributor to late deliveries, accounting for 41.60% of all late orders. First Class exhibited the highest late-delivery rate at 100%.

The Ishikawa diagram and 5 Whys analysis identified potential improvement opportunities involving shipping schedule calibration, operational variability, and performance monitoring. These potential causes require further investigation before causal conclusions can be established.

Three hypothetical improvement scenarios were modeled. The proposed target scenario reduced the late-delivery rate to 45%, corresponding to approximately 7,744 fewer late orders over an equivalent order volume.

The project demonstrates the application of DMAIC, Power Query, KPI analysis, PivotTables, Pareto analysis, root-cause investigation, scenario modeling, and control planning.

Limitations: The proposed improvements have not been implemented or experimentally validated. True Process Cycle Efficiency could not be calculated without defensible value-added processing-time measurements. The findings should therefore be interpreted as a historical analytical case study rather than verified operational improvement results.

Recommendation: Future work should validate the suspected causes of delivery delays, test scheduling and process improvements in a controlled operational environment and evaluate post-implementation performance before making claims of sustained improvement.

7. REFERENCES

DataCo Global. (n.d.). *DataCo Smart Supply Chain for Big Data Analysis* [Dataset]. Kaggle. https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

Microsoft. (n.d.). *Microsoft Excel and Power Query*. Microsoft Corporation.

Data Source Note: The analysis uses a publicly available historical supply chain dataset. All improvement scenarios are hypothetical and have not been operationally implemented.